Summary Notes:
Fundamentals of Cloud Computing

1. Introduction

Overview

Here is an overview of what was covered in the session:

1. Cloud Deployment Models
    - Public Cloud
    - Private Cloud
    - Hybrid Cloud
    - Multi-Cloud
    - Intercloud
    - Community Cloud
2. Cloud Security and Compliance
    - Shared Responsibility
    - Data Encryption
    - Collaborative Management
    - Security and Compliance Monitoring
3. Emerging Trends and Future of Cloud Computing
    - Current Trends
    - Innovations in Cloud Technology
    - Future Outlook and Predictions

2. Cloud Deployment Models

Cloud deployment models determine how cloud services are made available to users and how resources are managed and distributed. Several deployment models exist for the cloud: public, private, hybrid, multi-cloud, intercloud and community cloud. Each model grants users access to the applications and files at any time and from anywher, but they differ in their approach to resource management, security and control. The choice of cloud deployment model for your business depends on various factors, including the intended use of the cloud environment, regulatory requirements and specific business needs.

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2.1 Public Cloud

A cloud service provider hosts and distributes a public cloud via the internet. These deployment models are the most affordable and well-liked options, as they relieve clients of the burden of buying, overseeing, and maintaining their own IT infrastructure. Public cloud providers like Microsoft Azure, own, operate, and take full responsibility for the hardware, infrastructure, and data centres that house the workloads of their customers. Generally speaking, they offer high-bandwidth network connectivity to guarantee excellent performance and quick access to data and apps. The data centre infrastructure and other resources of the cloud provider are shared by all tenants in multi-tenant public cloud settings.

Example: **Microsoft Azure**

Microsoft Azure offers a robust public cloud platform that offers a range of services, including virtual machines, databases and artificial intelligence. These services are hosted by Microsoft and accessed over the open internet, relieving customers of the burden of purchasing, maintaining and managing their own IT infrastructure. Public cloud offerings from Azure include SaaS applications, individual virtual machines and enterprise-grade infrastructures. These resources may be available at no cost, or they may be billed on a pay-per-use or subscription basis.

2.2 Private Cloud

Computing resources devoted solely to an organisation make up a private cloud. It may be physically housed by a cloud provider or at the on-site data centre of an organisation. Customers of private clouds get more control and customization along with the main advantages of a public cloud, such as elasticity, scalability and self-service. Because private clouds are hosted on private networks that are inaccessible to the general public, they provide a better level of privacy and security. To comply with regulations, many businesses - particularly those in the public, medical, and financial sectors - opt for private cloud environments.

In order to comply with regulations, many businesses choose for a private cloud environment over a public one. When working with workloads involving sensitive data such as financial information, medical records, intellectual property, personally identifiable information (PII), or confidential documents, organisations like government agencies, healthcare providers, and financial institutions frequently choose to use private cloud environments.

Example: **Microsoft Azure Stack**

Microsoft Azure Stack allows organisations to deploy Azure services in their own data centres, creating a private cloud environment. By providing the business with specialised resources, this configuration offers a higher level of privacy and security.

2.3 Hybrid Cloud

Combining public and private clouds with technologies to allow data and apps to collaborate is known as a hybrid cloud. This architecture facilitates workload portability and automatic deployment across many environments by providing a cost-effective, secure, and adaptable environment. Compared to utilising a

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public or private cloud alone, it helps organisations to fulfil their technological and business goals more successfully and affordably. For situations like DevOps, where apps may be developed and tested on the public cloud and then transferred to a private cloud for production based on security considerations or business requirements, hybrid clouds are perfect.

Example: Microsoft Azure Hybrid Cloud Solutions

Hybrid cloud solutions from Microsoft Azure make it possible to integrate Azure with on-premises infrastructure. This enables businesses to host less sensitive, customer-facing endpoints in the public cloud and run sensitive apps and services in a secure private cloud.

2.4 Multi-Cloud

Multi-clouding is a cloud computing approach which leverages multiple cloud providers' best offerings to implement a solution. Data governance requirements, workload and business are the driving forces for this strategy. SaaS, IaaS and PaaS are integrated in a loosely or tightly coupled architecture in a multi-cloud system. Businesses use multi-cloud to access more services, promote innovation, and avoid vendor lock-in. For instance, a business may choose to employ IBM Cloud for its data analytics capabilities, AWS for its worldwide reach in web hosting, and Azure for its security features.

Example: Microsoft Azure Arc

Microsoft Azure Arc makes Azure management and services available to AWS and Google Cloud, enabling multi-cloud computing. With this strategy, businesses may take advantage of the finest offerings from several cloud service providers.

2.5 Intercloud

Intercloud is a cloud computing strategy that allows the active management of data across multiple cloud environments, with each service on a different cloud infrastructure. Intercloud strategies involve managing data egress charges and establishing architecture to use data across distinct cloud environments. This approach is important for integrating data and analytics workflows across different services and clouds, providing extra value by using different remote cloud environments and expert programs.

Intercloud allows you to take the insights that are hidden in your data and mine them for extra value by using different remote cloud environments – and the expert programs held within each – to approach data with a different, combination approach. So, if your business has, for instance, lots of unstructured data and you want to milk it for insights that can either shave percentage points off your outgoings or boost the value and the practical use of your data, there may well be value to you in the intercloud approach, rather than using a simple, iterative, multi-cloud environment to interrogate the data one cloud at a time.

Example: Microsoft Azure Intercloud Solutions

Azure supports intercloud connectivity through solutions such as Azure ExpressRoute, which enables secure connections between different cloud environments and on-premises infrastructure.

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2.6 Community Cloud

A community cloud is an infrastructure of cloud computing where several organisations pool services and resources according to shared regulatory and operational needs. Community clouds are owned, managed and operated by members of the community, third-party vendors or both. These clouds address the specific needs of industries with stringent regulations, such as healthcare, finance and education. Community clouds offer more security than public clouds but are more cost-effective than private clouds, making them suitable for growing organisations within regulated sectors.

Example: Microsoft Azure Government

Microsoft Azure Government provides a dedicated community cloud environment for U.S. government agencies and their partners, exemplifying a community cloud.

3. Cloud Security and Compliance

Historically, the greatest deterrent for businesses using cloud services—primarily public cloud services—has been security concerns. Compared to conventional IT settings, maintaining cloud security necessitates distinct protocols and personnel skill sets. Below is a discussion of some recommended practices for cloud security.

3.1 Shared Responsibility for Security

In general, the client is in charge of safeguarding its data in the cloud, and the cloud service provider is in charge of protecting the cloud's infrastructure. To guarantee complete protection, it is crucial to specify data ownership and security obligations between public and private third parties.

Example: Microsoft is in charge of protecting the network and physical data centres that make up Azure's cloud architecture. Within their Azure environment, customers are in charge of setting up network security, controlling user access, and safeguarding their data.

3.2 Data Encryption

While in use, in transit, and at rest, data should always be encrypted. To guarantee data integrity and confidentiality, customers must retain exclusive control over their hardware security modules and security keys.

Example: To manage encryption keys and secrets, Azure provides data encryption services like Azure Key Vault, Azure Storage Service Encryption, and Azure Disk Encryption.

3.3 Collaborative Management

Ensuring secure and seamless cloud integrations requires effective communication and well-defined, comprehensible procedures between the security, operations and IT teams. Tools for collaborative management facilitate these procedures and improve security in general.

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Example: Azure DevOps and Azure Active Directory (AAD) facilitate collaborative management by providing integrated tools for secure code deployment, user authentication and access management.

3.4 Security and Compliance Monitoring

Knowing all of the regulations and compliance requirements that apply to your industry is the first step towards security and compliance monitoring. To keep an eye on all data transfers across all environments—private, hybrid, edge, and on-premises—it is essential to set up active monitoring of all linked systems and cloud-based services. This continuous observation guarantees the continued efficacy of security protocols and the regular fulfillment of compliance obligations.

Example: Azure Security Center provides continuous security assessment, advanced threat detection and security recommendations. Azure Policy helps organisations enforce compliance with regulatory standards.

4. Emerging Trends and Future of Cloud Computing

4.1 Current Trends in Cloud Computing

Significant advancements in cloud computing in recent years have changed how businesses function and how people access digital resources.

4.1.1 General Advancements in Technology

Because of these developments, cloud services have become widely used. This is mainly because these technologies facilitate faster application deployment and more effective use of resources.

Example: Microsoft Azure continuously updates its services to leverage the latest technological advancements, such as integrating quantum computing capabilities with Azure Quantum.

4.1.2 More Adoptions for AI and ML

The cloud is the most economical way to get the massive data bandwidth and processing power that AI and ML platforms require. Together, these technologies enable developers and companies to build intelligent applications and make data-driven decisions. Optimising cloud migrations is a major factor in enhancing AI/ML capabilities.

Example: Azure Machine Learning and Azure Cognitive Services provide powerful AI and ML platforms that require significant processing power and bandwidth, which Azure efficiently delivers.

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4.1.3 More Focus on Security Enhancements

Improved cloud security features are offered by cloud service providers. The incorporation of enhanced security protocols, like identity management and encryption, has enhanced the security and confidence that businesses have in cloud computing.

Example: Azure Security Center increases corporate confidence by providing improved security capabilities including user authentication, identity management, encryption and compliance certifications.

4.1.4 Focus on Sustainability

Due to the enormous energy consumption of servers and data centres, businesses are investing in implementing energy-efficient solutions and renewable energy sources. Net-zero commitments have been made by major cloud providers to decrease their carbon footprints and assist clients in reducing the energy usage traditionally associated with on-premises configurations, making sustainability a crucial corporate governance responsibility.

Example: Microsoft has pledged to become carbon-negative by 2030 through the installation of energy-efficient technologies in their data centres and investments in renewable energy sources.

4.2 Innovations in Cloud Technology

4.2.1 Serverless Services

When a service or piece of code is serverless, it means that it doesn't require a dedicated virtual machine and can operate on demand for a brief period of time, typically in reaction to an event. Typically, serverless services have a pay-for-value pricing structure, automatic scalability, and integrated high availability.

In serverless computing, resources are allocated from a pool that is managed by the cloud provider, so you usually don't have to worry about the underlying server at all.

Example: Azure Functions provides a serverless compute service that automatically scales and charges only for the resources used, without the need for dedicated VM management.

4.2.2 Containers on Demand

Applications and dependencies are packaged into lightweight executable units of software called containers. All of the benefits of running virtual machines (VMs) are also present when using containers on demand, including lower costs and resource requirements.

The operating system kernel of the host computer is shared by containers. They can operate on a Kubernetes service or Docker Engine.

Example: Azure Kubernetes Service (AKS) allows for easy deployment and management of containerized applications, offering the advantages of VM efficiency with fewer resources and lower costs.

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4.2.3 Pre-Built Container Images

Docker containers and images enable rapid application deployment and scalability. Pre-built container images, such as those available in Azure Container Registry, can quickly bring up full applications and are vendor-agnostic.

An executable instance of a Docker image—which is determined by a Dockerfile—is referred to as a Docker container. An image's building instructions are contained in a Dockerfile, which is frequently based on another image. An Ubuntu image could serve as the basis for an image that contains the Apache HTTP Server, for instance. The Docker registry contains pre-defined Dockerfiles, but you can also create your own. Docker images can be used locally with your Docker installation or in any cloud that supports containers. Similar to pre-built virtual machine images, Dockerfiles are vendor-agnostic and may instantly launch an entire application.

Example: Azure Container Registry stores and manages container images, allowing for quick deployment and scalability of containerised applications.

4.2.4 Cloud GPUs, TPUs **and** FPGAs

GPUs, TPUs, and FPGAs that are accessible in the cloud may greatly speed up deep learning with huge data sets and models, making it simple to leverage these potent resources as needed.

On clusters of CPUs, deep learning with huge models and extraordinarily large data sets—which are necessary for accurate training - can frequently take considerably longer than a week. GPUs, TPUs and FPGAs may cut down training time dramatically, and having them available in the cloud makes it easy to employ them as needed.

Example: Azure provides specialised VM instances with GPUs, TPUs and FPGAs for high-performance computing tasks such as deep learning and large data set training.

4.3 Future Outlook and Predictions for the Cloud Industry

4.3.1 Blockchain

We anticipate further blockchain and cloud computing integration. Blockchain is becoming more accessible to individuals and organisations because of Blockchain-as-a-Service (BaaS) platforms like Azure Blockchain Service, which increase operational efficiency, security, and transparency.

Example: Azure Blockchain Service enables the creation, management and governance of blockchain networks at scale, making blockchain technologies more accessible to businesses.

4.3.2 Internet of Things (IoT)

The Internet of Things (IoT) is a network of physically linked objects that communicate, gather, and exchange data via software, embedded sensors, and network connectivity. These gadgets may exchange information and react to particular circumstances without the need for direct human intervention by

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using the internet. By advancing data processing, real-time analysis, storage, and AI/ML integration, cloud computing and IoT integration transforms data processing and utilisation across sectors.

Cloud systems foster innovation across industries with their unmatched scalability, enormous data processing capacity, and improved security.

Example: Azure IoT Hub and Azure Digital Twins provide comprehensive IoT solutions for connecting, monitoring and managing billions of IoT assets.

4.3.3 Edge Computing

In order to minimise latency and maximise bandwidth, edge computing processes data closer to its source rather than in a distant data centre.

Because of bandwidth restrictions, latency, and the sheer volume of data created, traditional cloud systems have shown to be unreliable. Nevertheless, edge computing is poised to revolutionise cloud computing in the future by:

- **Reduced latency**: By processing data at the point of generation, edge computing drastically shortens the path between IoT devices and the computational resources they require. Making decisions in real time is made possible by the reduced latency, which guarantees quicker reaction times.
- **Optimising bandwidth**: The vast volumes of data that Internet of Things devices send to the cloud have the potential to overburden network bandwidth. By using edge computing, data processing is ensured locally, and only relevant or condensed information is transferred to the cloud. This keeps network connections strong and speeds up data flow.

Edge computing is perfect for applications that need quick responses and effective data management since it allows real-time data processing, lowers latency, and maximises bandwidth utilisation.

Example: Azure IoT Edge brings cloud analytics and intelligence to edge devices, lowering latency and bandwidth consumption and facilitating real-time data processing.

6. Summary and Key Takeaways

1. Cloud Deployment Models
    - Public Cloud: Offers cost-effective, scalable and accessible resources managed by cloud service providers such as Microsoft Azure, Google Cloud and AWS. It's a multi-tenant environment where resources are shared among many users.
    - Private Cloud: Offers dedicated resources for a single organisation, ensuring higher security and privacy. Suitable for entities with regulatory compliance needs, such as government agencies and financial institutions.
    - Hybrid Cloud: Combines public and private clouds, enabling flexible deployment of applications and data. Ideal for organisations needing to balance security with scalability.

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- Multi-cloud: Utilises services from multiple cloud providers to avoid vendor lock-in and leverage the best features from each. Commonly used to enhance flexibility and innovation.
- Intercloud: Manages data across multiple cloud environments, enabling integrated data and analytics workflows. Suitable for businesses seeking to extract maximum value from diverse data sources.
- Community Cloud: Shared infrastructure among organisations with common regulatory and operational requirements, providing a balance between cost, security and compliance.

2. Cloud Security and Compliance
    - Shared Responsibility: The cloud provider secures the infrastructure, while customers protect their data.
    - Data Encryption: Essential for securing data at rest, in transit and in use.
    - Collaborative Management: Safe cloud integrations are ensured by efficient communication and well-defined procedures between the IT, operations, and security teams.
    - Security and Compliance Monitoring: To ensure data visibility and regulatory compliance, proactive monitoring of all linked systems and cloud services is performed.
3. Emerging Trends and Future of Cloud Computing
    - Current Trends:
        - Rapid adoption of cloud services due to advancements in technology.
        - Increased use of AI and ML, demanding high processing power and bandwidth.
        - Enhanced cloud security features building trust with businesses.
        - Focus on sustainability and renewable energy in cloud data centres.
    - Innovations in Cloud Technology:
        - Serverless services for efficient, on-demand computing.
        - Containers on demand for lightweight, cost-effective application deployment.
        - Pre-built container images for quick application setup.
        - Cloud GPUs, TPUs and FPGAs for faster training of deep learning models.
    - Future Outlook and Predictions:
        - Increased cloud computing and blockchain integration will increase security and operational efficiency.
        - Advancements in IoT, with cloud computing playing a crucial role in data processing and analysis.
        - Increased adoption of edge computing to reduce latency and optimise bandwidth, enhancing real-time decision-making capabilities.

By understanding these concepts and trends, participants will be well-equipped to navigate the evolving landscape of cloud computing, leveraging its benefits while addressing associated challenges.

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